

Design and Simulation of a Wideband GPR Antenna with FSS-MA Shielding

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Abstract—In order to improve the radiation bandwidth of the ground-coupled GPR antenna and the radiation performance in the preferred direction, a new wideband bowtie antenna with Frequency Selective Surface (FSS) shielding and Metamaterial Absorber (MA) is proposed in this paper. The proposed antenna composed of three elements: an original bowtie antenna working at the center frequency of 1GHz, a two-sided band-stop FSS coating, and a MA. The simulation results show that the proposed FSS-MA-bowtie antenna achieves a better performance in near-field radiation, VSWR and input impedance, comparing with the original bowtie antenna. The side radiation of the original bowtie antenna is efficiently suppressed by the band-stop FSS shielding and are absorbed by back reflection. Results show that the gain of the preferred (forward) radiation direction have increased by 4dB, and the bandwidth reaches 27% by employing the FSS-MA shielding mechanism. In addition, time-domain analysis results that reflected oscillation was reduced, which satisfies the requirements of a wideband GPR system.

I. INTRODUCTION

Non-destructive detection for pavement distress and subsurface fault has become an increasingly prominent need, with the rapid development of road traffic construction in China [1]. Ground Penetrating Radar (GPR) is very useful in detecting subsurface structures and objects such as cables, pipes hidden tunnels, and landmines [2]. As a wideband antenna, the bowtie antenna has advantage of satisfied bandwidth at a lower frequency, easy coupling with surface, and compact geometrical dimension. However, the electromagnetic wave radiation of the bowtie antenna is omnidirectional radiation [3], thus a GPR system using bowtie antennas has to employ a shielding/absorbing cavity to suppress the back and side radiation, which leads to a lower electromagnetic wave gain of its main radiation direction [4].

Metamaterials are artificial materials that can achieve properties that provide a new solution to this problem [5]. As an important direction of research for metamaterials, the Frequency Selective Surfaces (FSS) and Split-ring Resonator (SRR) have significant meaning in the Microwave field since they were proposed. It can be used as a shielding for GPR antenna, to control or alternate the electromagnetic waves as we preferred [6].

In this paper, a new bowtie antenna with a two-sided band-stop type FSS shielding and a back-sided Metamaterial

Absorber (MA) [7] for GPR applications is proposed. When compared with the original bowtie antenna, both the return loss and the gain of the proposed antenna is improved while the bandwidth is maintained.

II. FSS UNIT ANALYSIS AND DESIGN

The proposed split-ring resonator (SRR) unit is shown at the left of figure 1. The yellow part is SRR, the green part is FR4 dielectric substrate with a dielectric constant of 4.4, the thickness $t=1.5\text{mm}$. In this paper, we use electromagnetic simulation software Ansys HFSS v15 to calculate the characteristics of the shielding layer, using periodic boundary conditions and floquet Port to simulate infinite array. The simulation model is shown in the right of figure 1. The design is based on the perimeter size of $1/2 \lambda_g$, in the resonant frequency of 1GHz, which makes the outer edges of the dipole SRR about half a guide wavelength. The theory comes from Ben a. Munk's Frequency Selective Surfaces Theory and Design. On this basis, the SRR unit is imported into HFSS for optimization, and $c1$, $r1$, $r2$ is adjusted to make the surface of this frequency selection work on 1GHz.

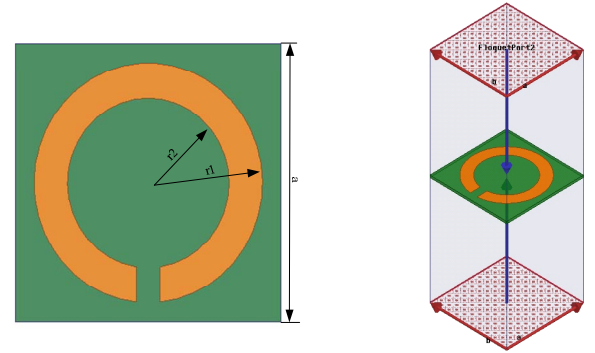


Figure 1. FSS unit and simulation setup

TABLE I

Ring FSS unit structure parameters

$c1/\text{mm}$	$r1/\text{mm}$	$r2/\text{mm}$	a/mm
8	24	17	55

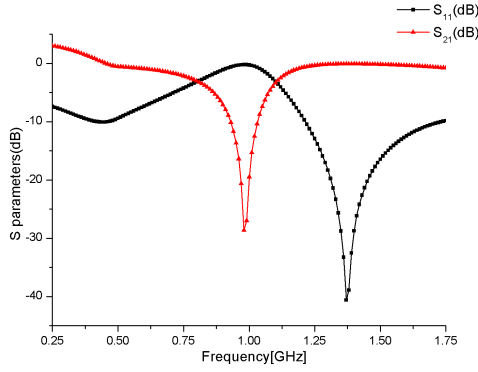


Figure 2. S parameter of FSS shielding

The resonant frequency is influenced by the parameters of the unit, and the perimeter of the unit determines unit resonant frequency. After a large number of simulation calculations, the parameters are obtained as shown in table I. Figure 2 shows the S parameter of FSS shielding after slight adjustment. When the circumference of SRR unit is $1/2 \lambda_g$, the resonant frequency is approximately 1GHz. At this operating frequency, the FSS shielding will reflect electromagnetic waves strongly.

III. ANALYSIS AND DESIGN OF THE BOW-TIE ANTENNA

The bowtie antenna has the characteristics of infinite length double-cone antenna impedance characteristic and frequency independence, and it can satisfy the design requirements of miniaturization of antenna. Figure 3 is a schematic of the bowtie antenna design. The length of the antenna size is mm, and its solution frequency is 1GHz. Table II shows the structural parameters of the bowtie antenna.

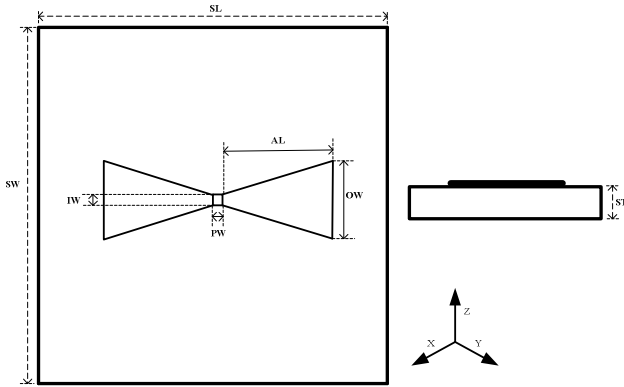


Figure 3. Schematics of bowtie-antenna design

TABLE II
Bowtie antenna structure parameters (unit: mm)

SW	SL	AL	PW	IW	OW	ST
167	167	45	2.08	2.08	37.48	1.06

IV .DESIGN AND SIMULATION OF BOWTIE ANTENNA WITH FSS SHIELDING AND METAMATERIAL ABSORBER

Simulation is conducted for the designed bowtie-antenna combined with the FSS shielding layer and MA. According to the direction of radiation pattern of bowtie-antenna, we place two finite periodic arrays of SRR-substrates off-center in the X-axis direction, which is arranged periodically. The space between of array unit space is 55 mm. The opening direction of SRR parallel to the Z-axis, so that it can inhibit lateral radiation of electromagnetic wave. In addition, a metamaterial absorber is placed on the high of 100mm above the bowtie antenna to absorb backward radiation, which towards the Z-axis negative direction to reduce the oscillation of backward electromagnetic wave and improve the vertical radiation gain. Figure 4 (a) shows the simulation model of top sheet of MA. The MA consisting of multiple metallic patches and a metallic ground plane separated by a single dielectric layer [7]. Figure 4 (b) shows the MA with boundary conditions, which is the perfect electric boundary on the y z-plane and perfect magnetic boundary the x z-plane. The electromagnetic wave incidents along z-axis. The optimal parameters are as follows: radius of the circular patch $r_1=39\text{mm}$, $r_2=36\text{mm}$, $r_3=37\text{mm}$, $r_4=38\text{mm}$, $r_5=40$, $r_6=41\text{mm}$, $r_7=42\text{mm}$. The calculation formula of absorption rate is $A(\omega) = 1 - T(\omega) - R(\omega)$ [8]. $T(\omega)$ is the transmission coefficient S_{21} and $R(\omega)$ is the reflection coefficient S_{11} . Due to the backboard is made of metal, the $T(\omega) = 0$. In order to obtain a wide absorption band, we pack seven distinct metallic unit cell. Each cell has one distinct resonant peak at a very narrow band, so we can broaden the absorption bandwidth by packing them closely by different the geometric parameters of the unit cell. The simulation result of absorption is shown in Figure 5.

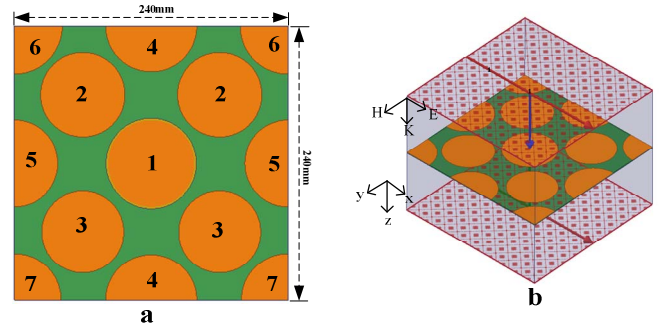


Figure 4. Top sheet of Metamaterial Absorber (MA)

- (a). Geometry of Metamaterial Absorber
(b). Setup for the numerical simulation

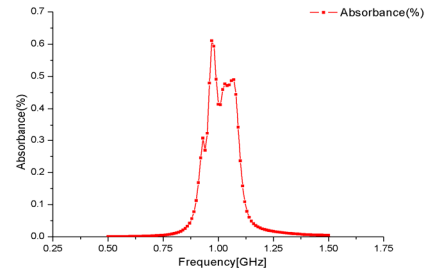


Figure 5. The absorbance of the top sheet for bowtie-antenna

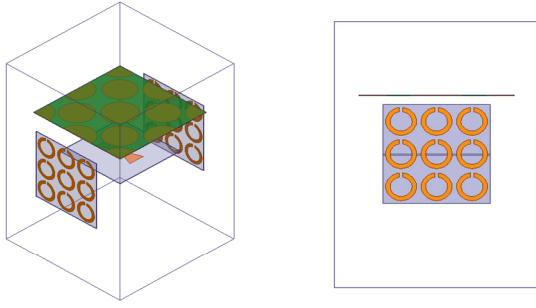


Figure 6. 3D view and side view of bowtie-antenna loaded with two-sided FSS shielding and top-sided metamaterial absorber

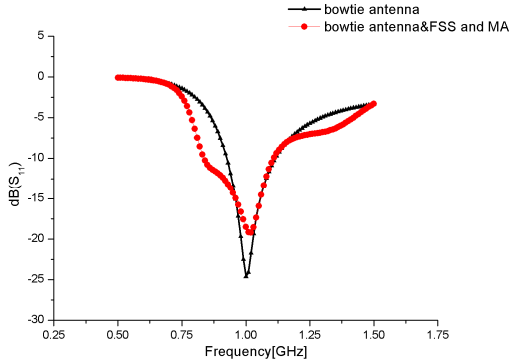


Figure 7. Return Loss of bowtie-antenna with FSS and MA compared with normal bowtie antenna

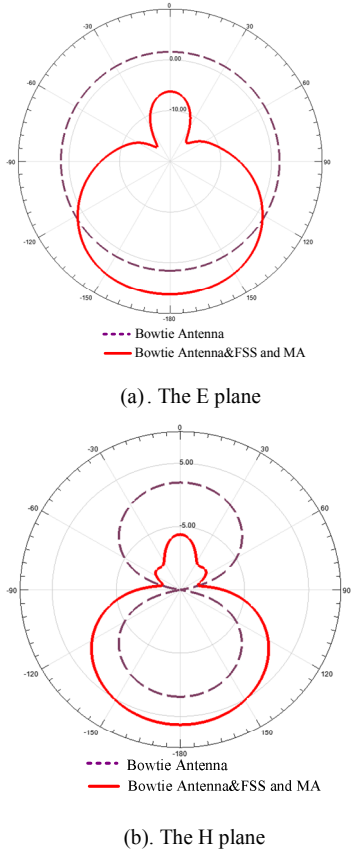


Figure 8. The E/H-plane of radiation pattern of the bowtie-antenna with FSS and MA shielding compared with normal bowtie antenna

Figure 6 and Figure 7 show 3D view of the bowtie antenna with FSS and MA shielding layer and the return loss of designed model, respectively. Figure 8(a) and Figure 8(b) show the E plane and H plane of radiation pattern. It can be seen from the figure that after loading the FSS shielding and metamaterial absorber, the impedance bandwidth increases to 27%, the gain of the preferred (forward) radiation direction is increased by 4dB, it can be concluded from the simulation results that loading the FSS shielding on the two sides and MA shielding on the top side can improve the gain in the preferred direction and thus improve the radiation efficiency of a ground-coupled GPR antenna. Figure 9 show the results that time domain analysis, it can be seen from the figure that the final model's reflected oscillation is reduced.

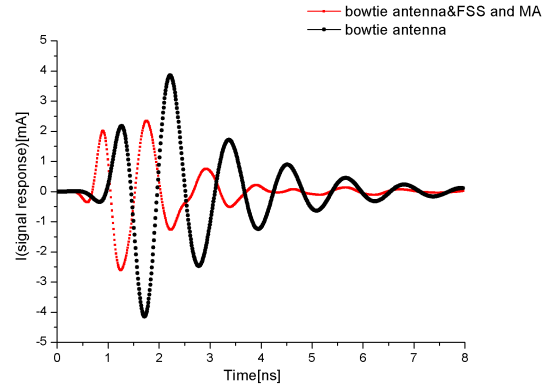


Figure 9. The time-domain analysis results of bowtie antenna with FSS and MA compared with normal bowtie antenna

V.CONCLUSION

This paper proposed a new wideband bowtie antenna with FSS-MA shielding, which design to improve the radiation performance in the preferred direction of an original bowtie antenna. By properly designing the reflective properties of the FSS and MA shielding, the side radiation of the original bowtie-antenna could be suppressed by the FSS shielding efficiently, while the back-reflected oscillation can be reduced by MA effectively. The simulation shows that with the FSS shielding, the gain of the preferred (forward) radiation direction is increased by 4dB; the impedance bandwidth of the antenna achieves 27%, and the time domain performance is improved significantly, which satisfies the requirements of wideband GPR systems.

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